



AI/ML & Backend Intern Assignment

Role: AI/ML & Backend Engineering Intern
Duration: 48 Hours from Assignment Receipt

Submission

Candidates are required to submit:

- A complete **GitHub repository**
- A **demo video (mandatory)** demonstrating the working system end-to-end

1. Objective

This assignment is designed to evaluate your ability to design and implement a **real-world intelligent system** that integrates:

- Applied **AI/ML concepts (specifically Retrieval-Augmented Generation)**
- Backend system design and API architecture
- Frontend interaction and user experience
- Data handling and system flow management

The expectation is not just correctness, but **clarity of thought, architectural decisions, and system design maturity**.

2. Problem Statement

You are required to build an **AI-powered role-based candidate screening system**.

The system should simulate a structured technical interview where questions are not predefined but instead are dynamically generated based on

- The candidate's **resume**
 - The **selected job role**
 - A **role-specific knowledge base (textbook/corpus)**
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3. Expected System Flow

- Your system should support the following flow:
- Candidate Entry
 - User uploads their resume (PDF or text)
 - User selects a target role (e.g., Backend Engineer, AI/ML Engineer)
- Resume Processing
 - The system parses the resume
 - Extracts relevant information such as:
 - Skills
 - Technologies
 - Domain exposure (if possible)
- Context Construction
 - Based on the selected role and extracted resume data:
 - Generate meaningful queries
 - Identify relevant topics or domains to evaluate
- Knowledge Retrieval (RAG)
 - Query a role-specific knowledge base
 - Retrieve the most relevant content chunks
 - Ensure the retrieved context is meaningful and grounded
- Question Generation
 - Generate interview questions using the retrieved context
 - Questions should:
 - Be relevant to the role
 - Be influenced by the candidate's background
 - Reflect conceptual and/or applied understanding
- Interactive Interview
 - Candidate answers questions through the UI
 - System maintains session continuity
 - Questions may adapt based on previous responses (optional but valuable)
- Response Handling

- Store candidate responses
 - Maintain structured records of:
 - Questions asked
 - Answers given
 - Final Output
 - Display a structured summary of the interaction
 - Provide basic insights or analysis of the session
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4. System Architecture Expectations

You are expected to build a **complete, modular system**, not a single script.

Frontend

- Implemented using React or Next.js
- Should provide:
 - Resume upload interface
 - Role selection
 - Interview interaction flow
 - Results or summary view

Backend

- Implemented using Python (FastAPI or Flask recommended)
- Should be responsible for:
 - Handling all business logic
 - Managing interview sessions
 - Orchestrating the AI/ML pipeline
 - Interfacing with storage systems

Data Layer

- A database must be used for persistence
- The system should store:
 - Session data
 - Questions and answers
 - Any generated outputs or reports

System Design

- Code should be modular and structured

- Separation of concerns is expected
 - Configuration should be handled through environment variables
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5. Backend & API Design Expectations

The backend should be designed as a **well-structured service layer** that exposes functionality required by the frontend.

Rather than focusing on specific endpoints, emphasis should be placed on:

- Clear separation of responsibilities
- Logical API design and request flow
- Proper handling of different stages of the interview lifecycle
- Robust error handling and validation
- Maintainable and scalable code structure

The system should reflect how a real-world backend would be designed to support such an application.

6. AI/ML Requirements (Core Focus)

This is a critical component of the assignment.

You are expected to implement a **Retrieval-Augmented Generation (RAG) pipeline** with clear reasoning behind your design choices.

7.1 Knowledge Ingestion

- Load and process role-specific documents
- Implement a chunking strategy
- Generate embeddings
- Store them in a vector database

Your choices here should reflect an understanding of:

- Context preservation
 - Retrieval efficiency
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7.2 Retrieval Mechanism

- Construct queries dynamically based on:
 - Resume content
 - Role selection
 - Retrieve relevant information from the knowledge base
 - Ensure retrieved content is actually useful for downstream tasks
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7.3 Question Generation

- Generate questions using the retrieved context
 - Avoid generic or template-driven outputs
 - Questions should reflect:
 - Depth
 - Relevance
 - Context awareness
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7.4 Resume Utilisation

- A resume should meaningfully influence the following:
 - Topic selection
 - Question difficulty
 - Direction of interview
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7.5 Output Structuring

- Maintain a structured pipeline:
 - Context → Question → Answer → Storage
 - Ensure traceability of how questions were generated
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7. Frontend Expectations

The frontend should not just be functional but should clearly demonstrate the following:

- Integration with backend services
- Smooth user interaction flow
- State handling across the interview process

The UI should make the system easy to use and clearly reflect the different stages of the interaction.

8. Creativity & Extensions

This assignment is intentionally open-ended in several areas.

Candidates are encouraged to introduce their own ideas to improve the system—whether in terms of:

- Question formats
- Interaction design
- System intelligence
- User experience
- Evaluation logic

Stronger submissions typically go beyond the baseline implementation and demonstrate thoughtful enhancements.

9. Knowledge Base Resources

Candidates are expected to use the provided book as the **primary knowledge source for RAG implementation**, not generic internet queries

AI / Machine Learning Role

Machine Learning — Tom Mitchell

[Download PDF](#)

The Hundred-Page Machine Learning Book — Andriy Burkov

[Download PDF](#)

Machine Learning for Absolute Beginners

[Download PDF](#)

Data Science / Applied ML Role

Introduction to Machine Learning with Python

[Download PDF](#)

Master Machine Learning Algorithms — Jason Brownlee

[Download PDF](#)

Advanced / Theoretical ML (Optional for Strong Candidates)

Pattern Recognition and Machine Learning — Christopher Bishop

[Download PDF](#)

Artificial Intelligence, Machine Learning & Deep Learning

[Download PDF](#)

10. Deliverables

Repository

- Clean and well-structured codebase
- Proper README including:
 - Setup instructions
 - System architecture
 - Key design decisions

Demo Video (Mandatory)

The video should clearly demonstrate:

- Complete system flow
 - Key features
 - How different components interact
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11. Final Note

This assignment is designed to assess your ability to:

- Build **integrated AI-driven systems**
- Think in terms of **architecture and data flow**
- Make **practical engineering decisions under constraints**

Clarity, structure, and reasoning will carry as much weight as implementation.